



Scenario 2: Reusing an RML Mapping

Artefact: **A YARRRML/RML mapping** used to generate an RDF knowledge graph.

Scenario:

Imagine that you are building a knowledge graph for a new railway data integration project. You find an existing mapping from the European Union Agency for Railways (ERA), where YARRRML rules are converted to RML and executed with RMLMapper to transform source data into RDF.

Your task:

Review the artefact and the two metadata representations shown below. Then decide whether, based on the metadata alone, you would consider this mapping reusable as a starting point for a similar mapping task in a new setting.

What you are evaluating:

You are not being asked to judge the full technical correctness of the mapping itself. You are being asked whether the metadata description gives you enough information to make an informed reuse decision.

Please note:

You do not need to inspect every mapping rule in detail. Please base your decision only on the metadata shown on this page and assume that detailed technical inspection is not available at this stage.

▼ Mapping file (excerpt)

```

prefixes: grel: "http://users.ugent.be/~bjdmeest/function/grel.ttl#" eu-country:
"http://publications.europa.eu/resource/authority/country/" rdfs: "http://www.w3.org/2000/01/rdf-schema#" era:
"http://data.europa.eu/949/" xsd: "http://www.w3.org/2001/XMLSchema#" variables: connection: access: &host ${RINF_HOST} type: &typeMSSQL
mssql credentials: &credentials username: ${RINF_USER} password: ${RINF_PWD} queryFormulation: &queryFormulation sql2008 mappings: line-
national-ids: sources: - access: *host type: *typeMSSQL credentials: *credentials queryFormulation: *queryFormulation query: SELECT
DISTINCT UICLines.UICCode, MemberStates.Code FROM ... s: http://data.europa.eu/949/functionalInfrastructure/nationalLines/$(UICCode) po: -
[a, era:NationalRailwayLine] - [rdfs:label, $(UICCode)] - p: era:inCountry o: - mapping: eu-country condition: function: equal parameters:
- [str1, $(Code)] - [str2, $(code)] graph: ${NAMED_GRAPH} eu-country: sources: - ["data/countries.csv~csv"] s: $(eu)

```

(Please use only the information provided on this page for your decision.)

Show metadata (Representation A: FAIR-IMPACT metadata model)**Show metadata (Representation B: MMV lifecycle-based metadata model)****Questions (Scenario 2)**

1) Based on the metadata shown, would you consider this artefact reusable as a starting point in a new project?

Representation A: ☐ Yes ☐ Possibly, but with reservations ☐ No

Representation B: ☐ Yes ☐ Possibly, but with reservations ☐ No

If you selected “Possibly, but with reservations”, please briefly explain your reservations (for either representation):

(Optional)

2) Which representation better supported your reuse decision?

- ☐ Representation A
- ☐ Representation B
- ☐ Both equally
- ☐ Neither

3) Which representation was easier to understand?

- ☐ Representation A
- ☐ Representation B
- ☐ Both equally
- ☐ Neither

4) Which representation gave you more confidence in your decision?

- ☐ Representation A
- ☐ Representation B
- ☐ Both equally
- ☐ Neither

5) Do you think the two metadata models complement each other?

- ☐ Yes, clearly
- ☐ Somewhat
- ☐ No, they mostly overlap
- ☐ No, they are not compatible
- ☐ Not sure

6) What information was most useful for making your reuse decision?

(Optional)

**7) What important information was missing?**

(Optional)

**8) Any additional comments?**

(Optional)



Submit and continue to Scenario 3 →

After submitting, you will be taken directly to Scenario 3.

Copyright 2026, Sarah Alzahrani salzahra@tcd.ie

